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LOCAL SEQUENCE ALIGNMENT OF DNA AND PROTEIN SEQUENCES USING SMITH-WATERMAN ALGORITHM

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ABSTRACT

This project aims to develop a Java-based bioinformatics application for comparing DNA and protein sequences and identifying the most similar local regions between them. The system will implement the Smith–Waterman algorithm to perform local sequence alignment by calculating alignment scores and identifying the optimal matching region between two input sequences. The application will allow users to enter or upload DNA or protein sequences, configure match, mismatch, and gap scores, and view the resulting aligned sequences, alignment score, and percentage similarity. The project will be developed using Java, with data structures such as two-dimensional arrays used to construct and process the alignment matrix. The application can be used for detecting conserved regions, studying mutations and sequence variations, comparing genes or proteins, and supporting biological sequence analysis. The final system will provide an interactive and easy-to-understand alignment report while demonstrating the practical application of sequence analysis in bioinformatics.

PROBLEM STATEMENT

Comparing DNA and protein sequences is an important task in bioinformatics for identifying similarities, mutations, conserved regions, and evolutionary relationships. Traditional sequence comparison methods may not effectively identify highly similar regions within larger sequences, especially when mutations, insertions, or deletions are present.

The proposed project aims to develop a Java-based local sequence alignment system using the Smith–Waterman algorithm. The system will accept two DNA or protein sequences as input, calculate the optimal local alignment using configurable match, mismatch, and gap scores, and display the best matching regions along with the alignment score and percentage similarity.

The system will provide a simple interface for users to enter sequences and analyze their similarities. It can be useful in gene comparison, protein analysis, mutation identification, conserved-region detection, and biological research. The project will demonstrate how sequence alignment can be implemented as a practical bioinformatics application using Java.

PROPOSED

FUNCTIONALITIES

Sequence Input – Allow users to enter or upload two DNA or protein sequences for comparison.

Sequence Validation – Check whether the entered sequences contain valid DNA or protein characters.

Local Sequence Alignment – Implement the Smith–Waterman algorithm to find the most similar regions between the two sequences.

Scoring Configuration – Allow users to set match, mismatch, and gap penalties according to their requirements.

Alignment Matrix Generation – Generate and display the dynamic programming matrix used to calculate the optimal local alignment.

Traceback and Alignment Generation – Trace the optimal path through the matrix and display the final aligned sequences.

Similarity Calculation – Calculate and display the alignment score, number of matches, mismatches, gaps, and percentage similarity.

DNA and Protein Support – Support both DNA sequences and protein sequences with appropriate scoring options.

Result Visualization – Present the aligned sequences and important results in a clear and user-friendly format.

Performance Analysis – Measure the execution time for different sequence lengths to observe the performance of the alignment process.

Java-Based Implementation – Develop the complete core system in Java, using arrays and other suitable data structures for sequence processing and alignment.

Alignment Report – Generate a final report containing the input sequences, optimal alignment, alignment score, similarity percentage, and other relevant results.

TOOLS & TECHNOLOGIES

TO BE USED

Category	Tools / Technologies	Purpose
Programming Language	Java	Develop the core application and implement the Smith–Waterman algorithm
Development Kit	JDK 17 or above	Compile and execute the Java application
IDE	IntelliJ IDEA / Eclipse / VS Code	Write, debug, and run the Java program
User Interface	JavaFX	Create an interactive interface for entering sequences and viewing results
Algorithm	Smith–Waterman Algorithm	Perform local alignment of DNA and protein sequences
Data Structures	2D Arrays, Strings, ArrayList	Store sequences, alignment matrix, and alignment results
File Handling	Java I/O	Read and process sequences from text or FASTA files
Database	MySQL	Store sequence data and alignment results
Database Connectivity	JDBC	Connect the Java application with MySQL
Version Control	Git & GitHub	Manage and maintain project source code
Testing	JUnit	Test sequence validation and alignment functionality

EXPECTED

OUTCOMES

The expected outcome of this project is to develop a Java-based application that can accurately compare DNA and protein sequences using the Smith–Waterman algorithm.

The system will identify the most similar local regions between two sequences and display the optimal alignment along with the alignment score and percentage similarity. It will also identify matches, mismatches, insertions, and deletions and allow users to configure scoring parameters such as match, mismatch, and gap penalties. The application will provide a simple and user-friendly interface for entering or uploading sequences and generating clear alignment results. Overall, the project is expected to provide a useful tool for sequence comparison, conserved-region identification, mutation analysis, and basic biological sequence research.

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